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3.1 Measurements and Their Errors

1. SI Base Units

The International System of Units (SI) defines seven base quantities, each with a base unit. These form the foundation of all physical measurement. Every other unit in physics can be derived from combinations of these seven.

Base Quantity	SI Base Unit	Symbol
Mass	kilogram	kg
Length	metre	m
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

Note that the kilogram is the only base unit with a prefix. The base unit of mass is **not** the gram. This matters when converting: 1 g = 10^{-3} kg, not 1 kg = 10^{-3} g.

2. Derived Units

Derived units are combinations of base units. They can always be expressed in terms of kg, m, s, A, K, mol. Being able to derive the base units of any quantity is essential for checking equations.

Quantity	Unit	Symbol	In Base Units

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Force	newton	N	kg m s^{-2}
Energy	joule	J	kg m^2 s^{-2}
Power	watt	W	kg m^2 s^{-3}
Pressure	pascal	Pa	kg m^{-1} s^{-2}
Charge	coulomb	C	A s
Potential difference	volt	V	kg m^2 s^{-3} A^{-1}
Resistance	ohm	Ω	kg m^2 s^{-3} A^{-2}
Frequency	hertz	Hz	s^{-1}

How to Derive Base Units

Start from a defining equation and substitute until only base units remain.

Worked Example: Show that the volt has base units kg m^2 s^{-3} A^{-1}.

V = W / Q (energy per unit charge)
W is in joules: J = N m = (kg m s^{-2})(m) = kg m^2 s^{-2}
Q is in coulombs: C = A s
So V = kg m^2 s^{-2} / (A s) = **kg m^2 s^{-3} A^{-1}**

Homogeneity of Units (Dimensional Analysis)

Homogeneity: An equation is homogeneous if the units on the left-hand side are identical to the units on the right-hand side. Every valid physics equation must be homogeneous, though a homogeneous equation is not necessarily correct (it could have a wrong numerical factor or missing terms).

Worked Example: Check whether $E_k = ½mv^2$ is homogeneous.

LHS: E_k is in joules = kg m^2 s^{-2}

RHS: ½ is dimensionless, m is in kg, v^2 is in $(m\ s^{-1})^2$ = m^2 s^{-2}

So RHS = kg × m^2 s^{-2} = **kg m^2 s^{-2}**

LHS = RHS, so the equation is homogeneous.

Exam Tip: Dimensional analysis can help you spot errors in your working or eliminate wrong answers in multiple choice. If the units don't match on both sides, the equation (or your rearrangement of it) must be wrong. However, it cannot detect missing dimensionless constants like ½ or 2π.

3. SI Prefixes

Prefixes allow us to express very large or very small quantities conveniently. You must be able to convert between prefixed and standard form rapidly in exam calculations.

Prefix	Symbol	Factor	Example
tera	T	10^{12}	1 THz = 10^{12} Hz (infrared frequencies)
giga	G	10^{9}	1 GW = 10^{9} W (power station output)
mega	M	10^{6}	1 MΩ = 10^{6} Ω (high resistance)
kilo	k	10^{3}	1 km = 10^{3} m
centi	c	10^{-2}	1 cm = 10^{-2} m
milli	m	10^{-3}	1 mA = 10^{-3} A
micro	μ	10^{-6}	1 μm = 10^{-6} m (cell biology scale)
nano	n	10^{-9}	1 nm = 10^{-9} m (visible light wavelengths)
pico	p	10^{-12}	1 pF = 10^{-12} F (small capacitors)
femto	f	10^{-15}	1 fm = 10^{-15} m (nuclear radius)

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Exam Tip: When converting squared or cubed units, remember to apply the power to the prefix too. For example, 1 cm^2 = $(10^{-2}\ m)^2$ = $10^{-4}\ m^2$, not $10^{-2}\ m^2$. Similarly, 1 mm^3 = $(10^{-3}\ m)^3$ = $10^{-9}\ m^3$. This is one of the most common unit conversion errors in exams.

Real-World Application: SI units and prefixes are essential in engineering and medicine. A dosimetry error in 2000 at a Panama hospital, where radiation doses were calculated incorrectly due to a unit conversion mistake, resulted in several patient deaths. In aerospace, the Mars Climate Orbiter was lost in 1999 because one engineering team used imperial units while another used SI units. These cases demonstrate why rigorous unit handling is critical.

4. Random and Systematic Errors

Random error: An unpredictable variation in measurements that causes readings to scatter about the true value. Each measurement is equally likely to be above or below the true value. Reduced by repeating measurements and averaging.

Systematic error: A consistent error in the same direction every time a measurement is taken. It shifts all readings by the same amount (or proportion) away from the true value. It is NOT reduced by averaging.

Feature	Random Error	Systematic Error
Effect on readings	Scatter about the true value	All shifted in the same direction
Effect on mean	Mean approaches true value as repeats increase	Mean is offset from true value regardless of repeats
How to detect	Spread in repeated readings	Compare with a known accurate value
How to reduce	Repeat and average; use more sensitive instruments	Recalibrate instruments; correct technique; check zero error

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Examples	Fluctuations from electrical noise, varying reaction times, slight positioning differences	Zero error on a balance, parallax from consistent wrong angle, miscalibrated instrument

Zero error: A specific type of systematic error where an instrument gives a non-zero reading when it should read zero. Before taking measurements, always check and record any zero error, then subtract it from all subsequent readings.

Real-world application: In medical physics, systematic errors in radiation dosimetry equipment could lead to patients receiving incorrect doses. Regular calibration against national standards is essential. Random errors in blood pressure measurements are reduced by taking multiple readings and averaging.

5. Precision, Accuracy, and Resolution

Precision: How close repeated measurements are to each other. High precision means low spread (small random error). Related to the resolution of the instrument.

Accuracy: How close a measurement is to the true value. High accuracy means low systematic error.

Resolution: The smallest change in a quantity that an instrument can detect. For example, a ruler with mm markings has a resolution of 1 mm; a micrometer has a resolution of 0.01 mm.

Repeatability: Whether the same person using the same equipment gets the same results.
Reproducibility: Whether different people using different equipment get the same results.

A measurement can be precise but not accurate (e.g. consistently reading 10.3 when the true value is 10.0) or accurate but not precise (e.g. readings scattered around the true value with large spread).

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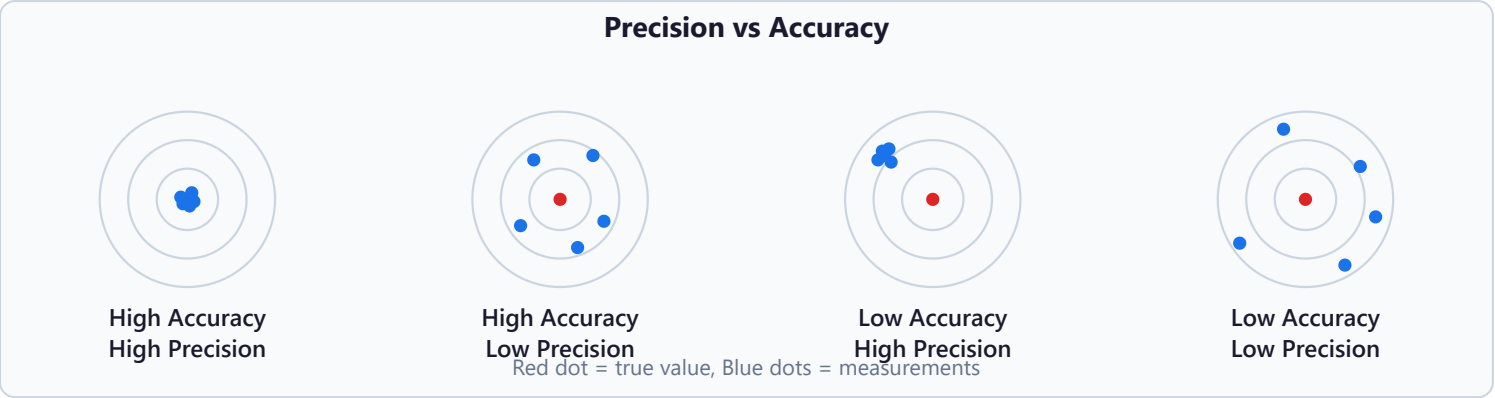


Exam Tip: Repeating measurements improves precision (reduces the effect of random error) but does NOT improve accuracy. To improve accuracy, you must eliminate systematic errors (e.g. recalibrate the instrument). These are different concepts — do not confuse them.

6. Uncertainties

6.1 Absolute Uncertainty

The absolute uncertainty is the margin of error in a measurement, expressed in the same units as the measurement.

- For a **single reading** with an analogue scale: uncertainty = ± half the smallest division.
- For a **single reading** with a digital instrument: uncertainty = ± the last digit (the resolution).
- For **repeated readings**: uncertainty = range / 2 (i.e. half the range of the data set).

Worked Example: A ruler with 1 mm divisions is used to measure a length. The reading is 245 mm.

Uncertainty = ± half the smallest division = ± 0.5 mm
Length = 245.0 ± 0.5 mm = 0.2450 ± 0.0005 m

Worked Example: Five measurements of a time period give: 2.31, 2.28, 2.34, 2.30, 2.32 s.

Mean = (2.31 + 2.28 + 2.34 + 2.30 + 2.32) / 5 = 2.310 s
Range = 2.34 - 2.28 = 0.06 s

Uncertainty = range / 2 = 0.03 s
Result: T = 2.31 ± 0.03 s

6.2 Percentage (Fractional) Uncertainty

$$\text{Percentage uncertainty} = (\text{absolute uncertainty} / \text{measured value}) \times 100\%$$

Worked Example: A diameter is measured as 5.30 ± 0.05 mm. Find the percentage uncertainty.

Percentage uncertainty = (0.05 / 5.30) × 100% = **0.94%**

6.3 Combining Uncertainties

Operation	Rule	Example
Addition or Subtraction (e.g. y = a + b or y = a − b)	Add the *absolute* uncertainties: $\Delta y = \Delta a + \Delta b$	If a = 5.0 ± 0.1 and b = 3.0 ± 0.2, then y = 8.0 ± 0.3 or y = 2.0 ± 0.3
Multiplication or Division (e.g. y = ab or y = a/b)	Add the *percentage* uncertainties: $\%\Delta y = \%\Delta a + \%\Delta b$	If %Δa = 2% and %Δb = 3%, then %Δy = 5%
Raising to a power (e.g. $y = a^n$)	Multiply the percentage uncertainty by the power: $\%\Delta y = n \times \%\Delta a$	If $y = r^2$ and %Δr = 1.5%, then %Δy = 2 × 1.5% = 3%

Worked Example: Calculate the resistivity and its uncertainty given:
R = 4.50 ± 0.05 Ω, L = 0.800 ± 0.001 m, d = 0.28 ± 0.01 mm

$A = \pi(d/2)^2 = \pi(0.14 \times 10^{-3})^2 = 6.16 \times 10^{-8}\ m^2$
$\rho = RA/L = (4.50)(6.16 \times 10^{-8}) / 0.800 = 3.47 \times 10^{-7}\ \Omega\ m$

Now for the uncertainty:
%ΔR = (0.05/4.50) × 100 = 1.11%

$\%\Delta L = (0.001/0.800) \times 100 = 0.125\%$

$\%\Delta d = (0.01/0.28) \times 100 = 3.57\%$

$\%\Delta A = 2 \times \%\Delta d = 2 \times 3.57\% = 7.14\%$ (because A depends on d^2)

$\%\Delta\rho = \%\Delta R + \%\Delta A + \%\Delta L = 1.11 + 7.14 + 0.125 =$ **8.4%**

$\Delta\rho = 0.084 \times 3.47 \times 10^{-7} = 0.29 \times 10^{-7}\ \Omega\ m$

$\rho = (3.5 \pm 0.3) \times 10^{-7}\ \Omega\ m$

Note: The diameter has the largest percentage uncertainty, so it dominates the final uncertainty.

Exam Tip: When subtracting two similar values (e.g. a temperature change of 3 °C from readings of 22 °C and 25 °C), the percentage uncertainty can become very large even though the absolute uncertainties are small. This is why it is better to measure a large change where possible. For example, measure the time for 10 oscillations rather than 1.

6.4 Uncertainty in a Mean

When you calculate a mean from repeated measurements:

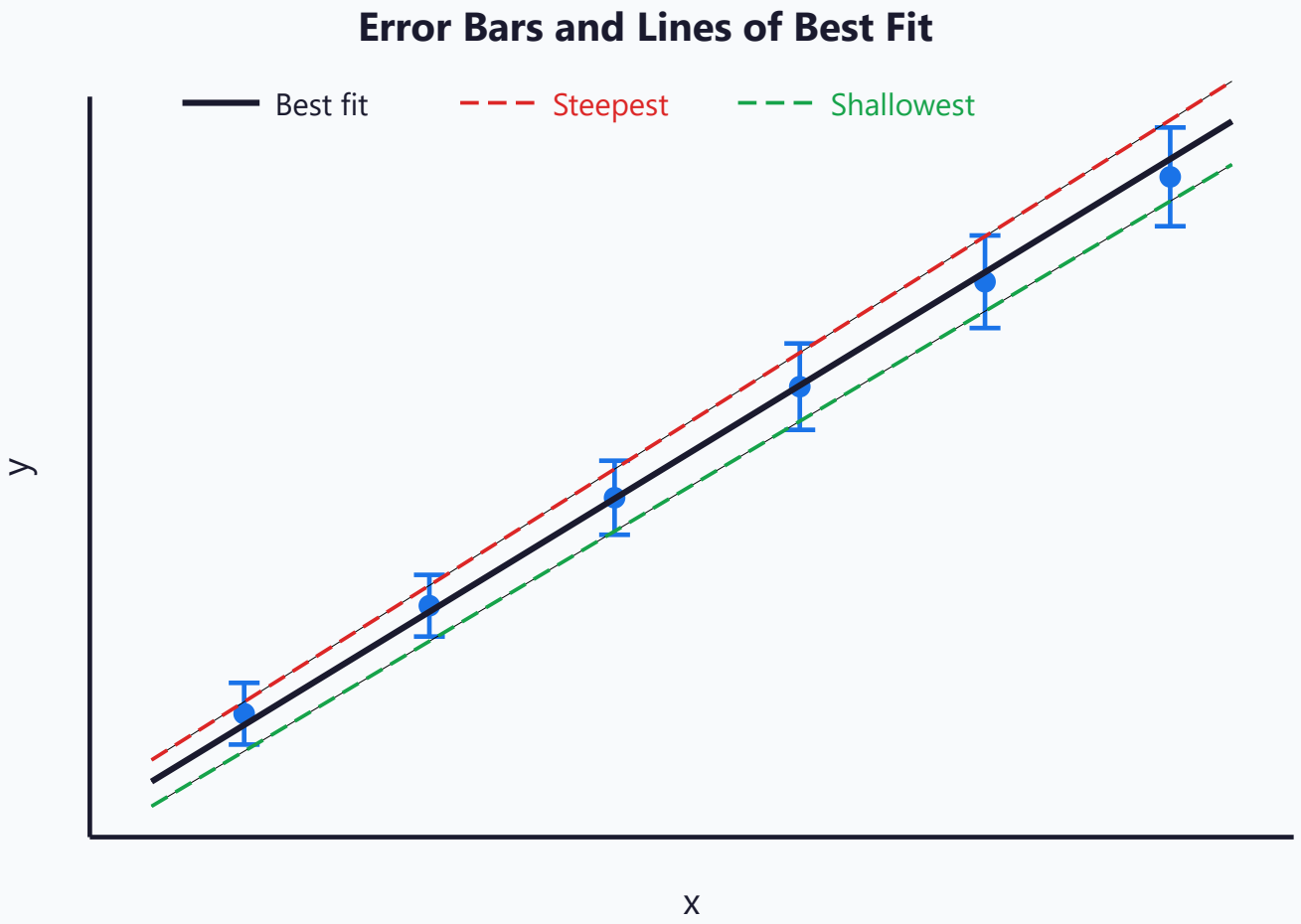
$$\text{Uncertainty in the mean} = \text{range} / 2$$

where range = maximum value − minimum value. If this is smaller than the instrument resolution, use the instrument resolution instead.

6.5 Uncertainties from Graphs

When determining a gradient from a straight-line graph:

1. Draw the **line of best fit** through the data points.
2. Draw the **steepest** acceptable line that still passes through the error bars.
3. Draw the **shallowest** acceptable line that still passes through the error bars.
4. Uncertainty in gradient = (steepest gradient − shallowest gradient) / 2.
5. The same approach applies to the y-intercept.



Worked Example: Reading gradients from a graph:

Line of best fit gradient = (260 - 48) / (430 - 80) = 0.606
Steepest gradient = (255 - 35) / (430 - 80) = 0.629
Shallowest gradient = (270 - 62) / (430 - 80) = 0.594

Uncertainty in gradient = (0.629 - 0.594) / 2 = ± 0.018
Gradient = 0.606 ± 0.018 (or 0.61 ± 0.02 to 2 s.f.)

Exam Tip: When plotting a graph with a y-intercept that has physical meaning (e.g. EMF from a V vs I graph), calculate the y-intercept uncertainty from the steepest and shallowest lines too. Use two large, well-separated points on each line to calculate the gradient — not the data points themselves.

7. Significant Figures

The number of significant figures in a result reflects the precision of the measurement.

- All non-zero digits are significant: 237 has 3 s.f.
- Zeros between non-zero digits are significant: 1002 has 4 s.f.
- Leading zeros are NOT significant: 0.0045 has 2 s.f.
- Trailing zeros after a decimal point ARE significant: 2.30 has 3 s.f.
- Trailing zeros without a decimal point are ambiguous: 1200 could be 2, 3, or 4 s.f. (use standard form to be clear: 1.2×10^3 for 2 s.f.)

Rules for Calculations

- **Multiplication/Division:** Round to the same number of significant figures as the value with the fewest s.f.
- **Addition/Subtraction:** Round to the same number of decimal places as the value with the fewest d.p.
- **Never round intermediate steps** — only round the final answer.

Worked Example: Calculate the kinetic energy of a 2.5 kg object moving at 3.42 m s^{-1}.

$E_k = ½mv^2 = ½ \times 2.5 \times (3.42)^2 = ½ \times 2.5 \times 11.6964 = 14.6205$ J

The value with fewest s.f. is 2.5 (2 s.f.), so E_k = **15 J** (2 s.f.)

Exam Tip: Your final answer should be given to the same number of significant figures as the data with the fewest significant figures used in the calculation. Do NOT round intermediate steps. If a question says "give your answer to 3 significant figures," follow that instruction even if the data would suggest otherwise.

8. Orders of Magnitude and Estimation

Order of magnitude: The power of ten closest to a quantity. Used for rough comparisons and sanity checks. Two quantities are of the same order of magnitude if one is no more than about 10 times larger than the other.

Common orders of magnitude you should know:

Quantity	Approximate Value	Order of Magnitude
Diameter of a nucleus	~1×10^{-15} m	10^{-15} m
Diameter of an atom	~1×10^{-10} m	10^{-10} m
Width of a human hair	~1×10^{-4} m	10^{-4} m
Height of a person	~2 m	10^{0} m
Diameter of the Earth	~1.3×10^{7} m	10^{7} m
Distance to the Sun	~1.5×10^{11} m	10^{11} m
Mass of an electron	9.1×10^{-31} kg	10^{-30} kg
Mass of a proton/neutron	1.67×10^{-27} kg	10^{-27} kg
Mass of a person	~70 kg	10^{2} kg
Mass of the Earth	~6×10^{24} kg	10^{25} kg
Speed of light	3×10^{8} m s^{-1}	10^{8} m s^{-1}
Speed of sound in air	~340 m s^{-1}	10^{2} m s^{-1}
Atmospheric pressure	~1×10^{5} Pa	10^{5} Pa

Estimation Strategy

When making an estimate:

1. Identify the relevant physics equation or principle.
2. Estimate each quantity to 1 significant figure or the nearest order of magnitude.
3. Perform the calculation, showing working clearly.
4. Give your answer to 1 s.f. or as an order of magnitude.
5. Sense-check your answer — does it seem reasonable?

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Worked Example: Estimate the number of atoms in a human body.

Mass of a person ≈ 70 kg
Human body is mostly water (H_2O), with molar mass ≈ 18 g/mol = 0.018 kg/mol
Number of moles ≈ 70 / 0.018 ≈ 4000 mol
Each molecule of water has 3 atoms
Number of atoms ≈ $3 \times 4000 \times 6 \times 10^{23} \approx 7 \times 10^{27}$
Order of magnitude: **10^{28} atoms**

Worked Example: Estimate the power output of a person running up stairs.

Mass ≈ 70 kg, height of one floor ≈ 3 m, time to climb ≈ 5 s
Work done = mgh = 70 × 10 × 3 = 2100 J
Power = W/t = 2100 / 5 ≈ **400 W** (order of magnitude 10^2–10^3 W)

Exam Tip: Estimation questions often appear on Paper 1 (worth 2-4 marks). They expect you to use sensible approximations and give your answer to 1 significant figure or as an order of magnitude. Show your working clearly — marks are given for your method even if your estimate differs from the mark scheme. Common estimation targets: number of molecules, energy costs, sizes of objects, forces in everyday situations.

9. Selecting Appropriate Instruments

Choosing the right measuring instrument depends on the quantity being measured and the precision required:

Instrument	Typical Resolution	Use For
Metre ruler	1 mm	Lengths from ~10 cm to 1 m
Vernier calipers	0.1 mm (or 0.05 mm)	Internal/external diameters, small lengths
Micrometer screw gauge	0.01 mm	Thin objects: wire diameter, thickness of a sheet

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Stopwatch (digital)	0.01 s	Time intervals (human reaction time adds ~0.2 s uncertainty)
Protractor	1°	Angles in optics experiments
Digital balance	0.01 g (or 0.001 g)	Mass measurements in the lab
Digital multimeter	Varies (0.01 V, 0.01 A typical)	Voltage, current, resistance
Oscilloscope	Depends on timebase setting	Rapidly varying voltages, frequency, waveform shape

Exam Tip: When asked to suggest improvements to an experiment, consider whether a more precise instrument could be used. For example, replacing a ruler (resolution 1 mm) with vernier calipers (resolution 0.1 mm) for measuring the diameter of a wire. Also consider whether timing multiple events (e.g. 10 oscillations) would reduce the percentage uncertainty.

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